

Project Proposal - GSoC 2026

Integrating HLS4ML with SOFIE for fast ML Inference
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Mentor Organization: CERN-HSF

Corresponding Project: ML4EP

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Personal Information of the Applicant

Brief Bio

Hi, I am Aditya Saroha final year student of Computer Science at the Visveswaraya Technological University. At the present I am working as a ML Intern at DRDO, conducting research based tasks under my supervisor(internship ending on April 30). I am fascinated by the power of AI since 2022, as it can literally have a mathematical relationship between anything which we can imagine and also by the strange maths it makes into use to solve a problem, I think we are close enough to solve the thinking problem analytically. Apart from that I am a very funny person who loves to do standup sometimes.

Previous Projects

I have made some projects in the domain of machine learning, web development etc. Some projects relevant to this tasks and other interesting things I have made are listed here: [Collision Simulation in 2D](#), [Interesting PINNs](#), [Evaluation Task](#)

Why ML4EP & CERN-HSF

So, it all started when I saw an internship post on linkedin about CERN summer student programme for the summer of 2023. I researched about the internship, read online about the tips and tricks for my application, I asked letter of recommendations from my professors and finally submitted the application. A few months later, I got the rejection email. At first I felt bad about it but afterwards I decided to give it another try next year and at the same time I saw the opportunity to contribute to open source projects related to CERN and I want to contribute my little part in discovering the next Higgs Boson.

Previous Contributions

I did not made any contributions specifically in the SOFIE architecture but I do have some small contributions in the Root software:

1. *Merged* [\[RDF\] Add option to change default basket size in RDataFrame Snapshot](#)
2. *Merged* [\[TFormula\] Pol Functions do not accept variable name as arguments](#)
3. *Merged* [\[RNTuple\] Improved RNTupleInspector::PrintColumnTypes\(\)](#)

4. *Merged* [Add bool to allow conversion between numpy arrays and RVec](#)
5. *Discussion* [TTree example is missing the critical lines](#)
6. *Open* [Initial changes for different snapshot behaviours in distRDF](#)

Contact Information

Here are my contact information: [LinkedIn](#) [Github](#) [Mobile](#) [Gmail](#)

Project Overview

Abstract

FPGAs i.e. Field Programmable Gate Arrays are semiconductor devices that are reprogrammable and allow for faster processing via parallel computation thus making themselves better candidates for edge AI applications. Many of these FPGAs are being used in CERN detector tunnels to help capture relevant data as each run of the detector may produce data in the order of Petabytes every second and storing it is not technically feasible. Now mostly all of these FPGAs for inferencing use the HLS4ML python package to create firmware level implementations of the machine learning models / neural networks that are trained using various other frameworks like pytorch or tensorflow. As these are very small devices hence comes with a tradeoff of storage and computation as well, FPGAs use various techniques like quantisation, compression, precision for efficient computation apart from that these devices must have high latency and higher throughputs. So, once the relevant data is stored via FPGAs inferencing within the detector, researchers would like to know on what basis FPGA classified this particular data to be worthy of storing hence we need a capability within SOFIE which can parse the in memory HLS4ML model with correct **datatypes**, correct **layers** and their **functions** in order to emit inference code in C++ that can be then run on CPUs for inferencing which then can be used for further study and improvement of the system.

Proposed Deliverables

- **HLS4ML integration** into the SOFIE architecture which includes the parser for the in memory model object of HLS4ML which will store the necessary information about the model configuration into a python dictionary which will then be used to instantiate the SOFIEs RModel object via the python interface. The layers of HLS4ML which will be supported by the parser to instantiate a

RModel object are listed below, which will support all of the associated SOFIE operators related to the layer(more layers can be added into this list during the coding period after discussing with the mentors): *Activation, Constant, Dense, Matmul, ELU, Reshape, Concatenate, Conv1D, Conv2D*.

- **Documentation** for Parser will have detailed APIs, Parser design, HLS4ML concepts & any other information to help new user adaptation. Tutorials for new users/learners & a test to check the developed Parser also will be added.

Plan of Work

I will start by going through the SOFIE architecture and its codebase in detail during the community bonding period and will also go through the HLS4ML codebase in order to understand its internals in more detail and all of the classes along with there associated functions. I will also study the SOFIE's keras parser which will give me a starting point to develop the parser for HLS4ML. Also during the community bonding period I will research on how we can combine HLS4ML with SOFIE in an efficient way with minimal exceptions as the HLS4ML framework is still under development. I will establish direct contact with the mentors, we will then again go through the finalized deliverables and will discuss further additions to the project goals and deliverables if necessary and also to the timeline of the project.

Understanding the Codebase, Research and Evaluation

Expected due date: May 24, 2026

Objective: To understand the internals of the SOFIE architecture, understand the codebase of the relevant parsers like SOFIEs parser for keras, HLS4ML and details about its backends and to find out ways in which we can integrate HLS4ML with SOFIE.

Deliverables:

- Understanding the SOIFEs architecture end to end and familiarisation with SOFIEs codebase.
- Understanding the architecture of other parsers like SOFIEs parser for keras.
- Familiarization with HLS4ML and its internals.
- Finding ways in which we can integrate HLS4ML with SOFIE.

I will start by understanding the architecture of the SOFIE by going through the code base [here](#), I will also dive deeper into the [internals of HLS4ML](#) and its [backend](#) and its [layers](#), its associated classes, functions etc. Apart from this I will also make myself familiarized with a new [keras parser](#) for the SOFIE implemented in python which

will give me a broad idea for the development of the parser. At the same time I will also understand the pytorch and ONNX [parsers](#) of the SOFIE which are based in C++ unlike the Python based implementation of the keras parser. This will help me to develop the parser either in C++ or in python.

I will research on ways in which we can integrate HLS4ML with SOFIE, as in the evaluation task I have added some pre-checks to the model being parsed, I will also research on these pre-checks and will add more of these for supported layers. One of the most important things in this period which will help me in the research is the interaction with my mentors, the discussion with the mentors and their direction will help me more than my individual research. So I will take directions from my mentors, discuss my doubts and any new frameworks which can be used in the project. The mentor's decision and direction will be final in carrying out the research and evaluation, if I have any new framework/direction I will discuss with the mentors before moving forward as shooting directionless in the sky will not find me the moon.

Phase 1 - Development of the Parser

Expected due date: July 01, 2026

Objective: To develop the initial parser which is capable of parsing the in memory HLS4ML model object and instantiate the RModel object via the python interface, the HLS4ML layers supported in this phase will be Activation, Matmul, Dense, Constant, ELU and any other layers which will be added into the project deliverables during the community bonding period after discussing with the mentors(as a whole 50% of the layers will be covered in this phase). The relevant tests, documentation and validation of the parser will also be covered.

Deliverables:

- Initial version of the standalone parser for HLS4ML capable of instantiating RModel objects.
- Relevant tests for the developed parser.
- Documentation for the parser.
- Validation of the inference results i.e. inferencing via HLS4ML and inferencing via SOIFEs emitted C++ code will be compared.

I have already developed a parser during the evaluation exercise, so I will take inspiration from that and I will also take inspiration from the

/tmva/sofie_parsers/src/RModelParser_PyTorch.cxx and

bindings/pyroot/pythonizations/python/ROOT/_pythonization/_tmva/_sofie/_parser/_keras/parser.py. Apart from that I will also discuss with the mentors on how I can integrate the pre-checks within the parser. During the coding period I will

simultaneously maintain the docs via in line comments and relevant doxygen docs if necessary, the same goes for the testing and validation of the parser. Detailed *project specific information*, *overview* and *pseudo code* of my implementation is available on the github gist [here](#).

Phase 2 - Development of the Parser

Expected due date: August 01, 2026

Objective: Implementation of the remaining layers of the HLS4ML into the parser which can instantiate a RModel object. Complete documentation of the parser along with tests.

Deliverables:

- A standalone parser for the HLS4ML capable of instantiating RModel objects with support for all of the layers described in the proposed deliverables section.
- Complete documentation and testing of the parser.

Rigorous testing and optimization

Expected due date: August 15, 2026

Objective: During this 15 days period, I will be completely dedicated to testing the parser, handling exceptional test cases and the optimization of the parser and the project as a whole.

Deliverables:

- Rigorous testing along with creation of specific tests.
- Optimizing the codebase and re-checking the validation of the inference.

I plan to document my coding journey through *github pages*, where I will add all the relevant information regarding the project like frameworks, methodology, etc.

Timeline

This is tentative and respect to change upon discussion and approval by mentors.

Days	Plan of Work
Community Bonding Period	
May 1 - May 24	• Understand the SOFIE architecture • Familiarisation with HLS4ML • Understand various parsers of SOFIE • Research ways to integrate HLS4ML into SOFIE

May 25 - July 1	● Initial version of the parser ● Parser capable of instantiating RModel object ● Tests, Documentation and validation of the parser ● Supports 50% of the layers in the finalized project deliverables ● Midterm evaluation report (1 Week buffer for mentor evaluation of the midterm report)
Midterm Evaluation (July 10)	
July 11 - August 1	● Parser development complete for the layers ● Documentation of the parser ● Tests for the new parser ● Start making the final evaluation report
August 2 - August 15	● Rigorous testing of the parser apart from phase wise tests ● Optimizing the codebase of the parser ● Handling exceptions (for pre-checks/complex models) ● Validation of the parser (1 week buffer for mentor evaluation of the final report)
Final Evaluation (August 24)	

Commitments

During this summer I am free and will be available for the entire duration of the project. I will commit 8 hours every day and 10 hours on weekends for this project which very easily allows me to satisfy the length of the [project](#). I check my email frequently and my response can be expected within 1 hour.

Post GSoC

Post the GSoC period, I will be continuing contributing to this amazing project and implementing more features requests for the new HLS4ML parser that may have not been covered in GSoC in the short time interval. I will also try to provide consistent patches and minor bug fixes. I am truly grateful to get the opportunity to contribute to this amazing project which is at the forefront of technical innovation.